

CS3520 Computer Organization and Architecture I

Domain Analysis Report

1. Application Context

The chosen application domain is low-cost AI-enabled mobile phones for the African context, with a specific focus on Lesotho. This domain addresses critical user needs such as enhancing digital security and accessibility, promoting financial inclusion through mobile money, and improving access to essential services in health and education. In regions with limited connectivity and resources, on-device AI processing is crucial. It enables functionalities like voice commands in local languages for users with low literacy, secure biometric authentication for mobile banking, and basic diagnostic tools that operate offline.

This domain is highly relevant as mobile phones are the primary gateway to digital services for many Africans. Designing processors that can run these AI applications efficiently and affordably is key to ensuring that the benefits of the Fourth Industrial Revolution are inclusive and tailored to local realities, rather than creating a dependency on expensive, imported technology.

2. Representative Applications

Application	Short Description	Key Computational Tasks	Notes(Relevance)
BIOMETRIC SECURITY	Provides secure device unlocking and transaction authorization using facial or fingerprint recognition.	Image preprocessing (edge detection), feature vector comparison, lightweight Convolutional Neural Network (CNN) inference.	Essential for building trust in mobile financial services and securing personal data on shared or low-cost devices.
MOBILE FINANCIAL SERVICES	Facilitates mobile banking, payments, and offline fraud detection.	Anomaly detection using small classification models, secure cryptographic operations (hashing, encryption).	Directly supports financial inclusion, a cornerstone of socio-economic development in Lesotho and similar economies.
VOICE AUTHENTICATION	Enables hands-free operation and voice assistants in local languages (e.g., Sesotho).	Real-time audio signal processing (filtering, FFT), feature extraction, execution of small Recurrent Neural Networks (RNNs).	Critical for accessibility and user interaction in a population with diverse literacy levels. Reduces reliance on text-based input.
HEALTH AI	Offers basic diagnostic aids, language translation, and	Execution of classification models (e.g., for symptom checking), simple	Addresses gaps in access to expert knowledge in rural and

	agricultural advisory tools.	Natural Language Processing (NLP) tasks.	underserved communities.

3. Workload Characteristics

The analyzed workloads are characterized by a mix of predictable, streaming data processing and more irregular, decision-based computations. There is a significant emphasis on integer and fixed-point arithmetic for signal processing and basic logic, with a secondary need for floating-point operations for accurate model inference in AI tasks. Data movement is a key factor: voice and image processing involve moving small-to-medium buffers of sensor data sequentially, while AI inference involves frequent access to pre-stored model weights.

Memory access patterns vary from sequential (audio buffers) to more random (accessing different parts of a neural network). Overall, the workloads require low-to-moderate computational intensity but must be executed with high efficiency and low latency to ensure a responsive user experience. Power consumption must be minimal to suit devices with limited battery capacity.

4. Initial Insights for Processor Design

- The processor requires efficient support for vector and matrix operations on small data sets, common in voice and image processing.
- Hardware support for multiply-accumulate (MAC) operations is critical for accelerating the inner loops of neural network inference.
- The ISA should facilitate efficient branching and conditional execution for the decision-making logic in fraud detection and classification tasks.
- A memory hierarchy capable of handling streaming data (for sensors) and providing low-latency access to small, critical data sets (like model parameters) is necessary.

5. References

- GSMA Intelligence. "The Mobile Economy Sub-Saharan Africa 2024."
- RISC-V International. (n.d.). RISC-V Specification Documents. Retrieved from <https://riscv.org/technical/specifications/> (The primary source for the base ISA).